

Python HW

You have one hour to work on this homework. Find one uninterrupted hour

- In a markdown cell write down what time you started the homework
- work for one hour or until done using your notes and other sources but not AI.
- After one hour ask AI to do each question.
- paste the AI answer in below your answer and run it also.
- you should have two clearly delineated code cells for each question with two sets of answers
- in a markdown cell below the two answers comment on the answers, differences, and what you have learned. (2-4 sentences)
- Your homework should look similar to this notebook. The only difference is your code won't be hidden. I have greyed out my code.
- Grading is not based on how well you coded but on how well you learned to code.

Start Time:

HW Part 1

Write 50-100 letters of nonsense in a string. But in that nonsense have a secret code that spells your name with every x letter (e.g. every 5th letter).

- print the whole string
- use slicing to print your name
- use slicing to print your name and then make it upper case

My Answer

- ```
a: My secret coded string!
Boerrianion4an6nn09 i Jkaj.hif ksfMifnasgmi30flbn-lniao238uasdx

b: My name in the string using slicing!
Brian J. Mailloux

c: My name in the string all in upper case
BRIAN J. MAILLOUX
```

## AI Answer

### Comparison

## HW Part 2

Repeat the same thing with a list but only do your first name. This is different as you are using a list. #2 above uses a string and this uses a list. It is a subtle difference but make sure to do it correct for each. Making a list is more annoying. So you only need to do your first name.

- print the list.
- Use slicing to print your name
- Print your name in reverse using slicing (for this to easily work the last letter of the list needs to be the last letter of your name)
- reverse the list
- print your name backwards using slicing on the reversed list.
- with the list reversed print your name forward with slicing.

## MY Answer

```
a: The list
 ['B', 'o', 'e', 'r', 'r', 'i', 'a', 'n', 'i', 'o', 'n', '4', 'a', 'n', '6',
 'n', 'n']
b: my name using slicing
 ['B', 'r', 'i', 'a', 'n']
c: my name backwards using slicing
 ['n', 'a', 'i', 'r', 'B']
d: I rverseed the list
 ['n', 'n', '6', 'n', 'a', '4', 'n', 'o', 'i', 'n', 'a', 'i', 'r', 'r', 'e',
 'o', 'B']
e: my name backwards using slicing
 ['n', 'a', 'i', 'r', 'B']
f: my name using slicing
 ['B', 'r', 'i', 'a', 'n']
```

## AI Answer

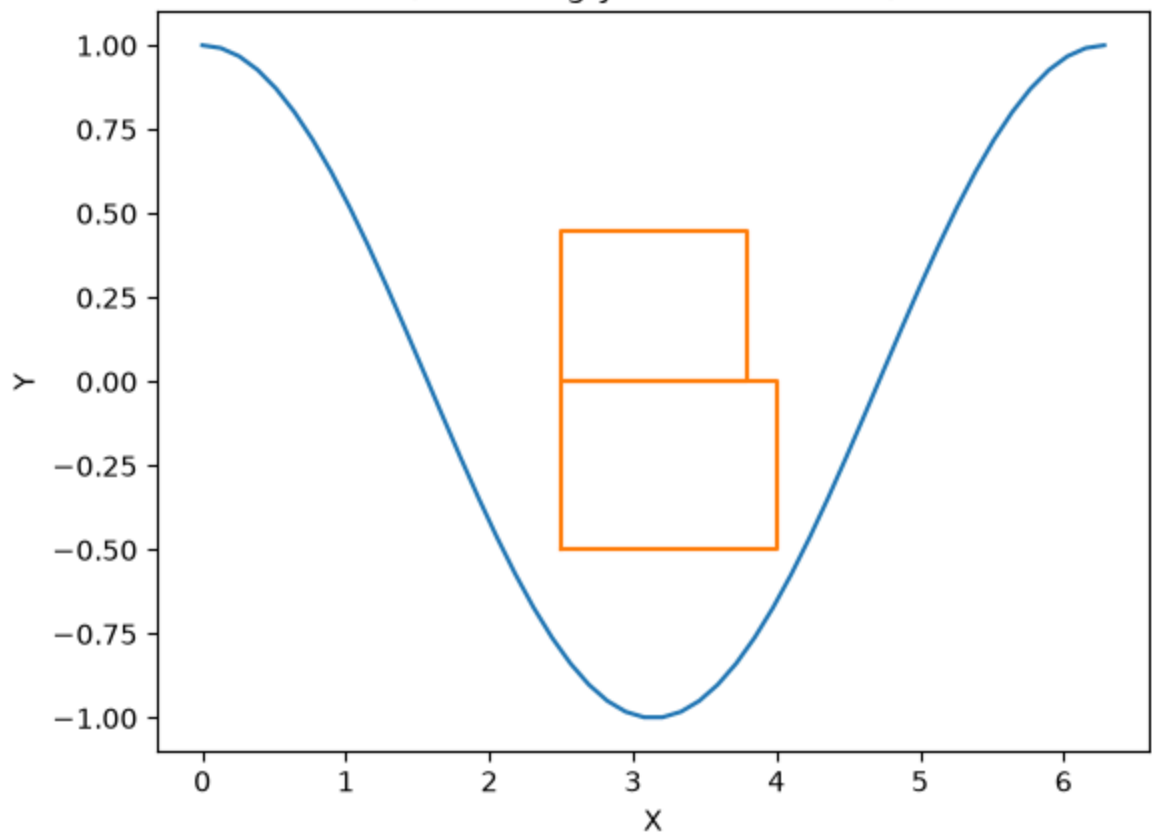
## Comparison

## HW Part 3

Final 10% today. Practice some more plotting and make the graph at the bottom of the page. Your graph should be your first initial. Not a B. You should be able to put together your array work with line plotting and cosine plotting to make it. The graph below shows cosine. I am okay with an ugly letter but it should be legible. You will need to create two and only two numpy arrays to make your letter if you set all the different coordinates correctly. Think back to your lists and line to think about how to make your letter.

## My Answer

B inside Cosine of X made using  
linspace, np.array([]) and ax.plot  
and a few other functions  
(It is an ugly B but that is ok)



AI Answer

Comparison